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DEEP LEARNING

Data Challenge Report

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Influencer or Observer: Predicting Social Roles

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Introduction

The objective of the challenge is to classify tweets as authored by either an Influencer (label 1) or an Observer (label 0). Each prediction must be made from a single tweet, making the task difficult due to the limited context. The dataset contains 154,914 training tweets and 103,380 test tweets. Simple baselines such as the Dummy Classifier or TF–IDF Logistic Regression show that raw lexical information alone is insufficient to capture the behavioural distinction between Influencers and Observers. To address this, we designed a hybrid high-performance pipeline combining: feature engineering, RoBERTa embeddings reduced by PCA, TF-IDF followed by a SVD, user groups learned via K-Means clustering, regularized ensemble models (LightGBM, XGBoost, CatBoost) and safe pseudo-labelling.

This pipeline achieved a final leaderboard score of 0.843 accuracy.

1 Exploratory Phase and Theoretical Background

Before establishing our final pipeline, we explored two distinct modelling paths to understand the complexity of the dataset.

1.1 Bridging Structure and Semantics: A Hybrid XGBoost Architecture

To address the challenge's goal of distinguishing social roles based on both structural and linguistic cues, we developed a hybrid architecture that fuses handcrafted statistical features with deep semantic representations. Our hypothesis was that while metadata captures the "high-leverage, asymmetrical" nature of Influencers, the nuance of their communication style required more sophisticated modeling than simple frequency vectors.

Feature Engineering and Extraction: We first constructed a robust set of 194 numerical features derived from the user and tweet metadata. Central to this was the implementation of "network ratios" (e.g., log(followers/friends) and followers-per-tweet), which directly model the asymmetry inherent in influencer profiles.

Contextual Embeddings (Twitter-RoBERTa): For the textual component, we moved beyond standard TF-IDF to capture semantic context. We utilized Twitter-RoBERTa-base, a transformer model pre-trained on around 58 million tweets. Unlike standard BERT, this model is adapted to handle social media patterns such as handles, emojis, and hashtags. We processed the tweets through the transformer and applied mean pooling to the last hidden state to generate a 768-dimensions embedding vector for each tweet.

Model Architecture and Results: We concatenated the statistical features and the transformer embeddings into a single feature space. This high-dimensional input was fed into an XGBoost classifier , chosen for its ability to handle tabular data and prevent overfitting via regularization (L1/L2). This "Feature Fusion" approach achieved a significant performance milestone, reaching an accuracy of 83.9% on the Kaggle leaderboard. This experiment validated that while network metrics are highly predictive, augmenting them with deep linguistic features provides a substantial gain over the logistic regression baseline, though it also highlighted the computational cost of generating embeddings for large datasets.

2 Final pipeline : Architecture and Strategy

2.1 Data Preprocessing and Feature Engineering

2.1.1 User and Tweet Behaviour Features

Due to the social nature of the task, we engineered more than 70 behavioural features, including:

- log-scaled follower/friend counts and ratios,
- engagement statistics (retweets, favourites, replies),
- account age, tweet frequency, activity rate,
- profile completeness (description, URL, location, banner),
- device source (web / Android / iPhone),
- media, hashtag, mention, and URL counts.

Moreover, we applied keyword detectors to user biographies (business inquiries, PR, social links), which are frequently indicative of Influencer accounts.

2.1.2 Text Representation

For each sample, we concatenated the content of each tweet with the description of the user's profile to take into account these descriptions in our prediction.

We extracted two complementary vector representations:

- RoBERTa embeddings: we used the `cardiffnlp/twitter-roberta-base-sentiment-latest` model to convert each text into a 768-dimensional vector using the CLS token, capturing the overall meaning of the sentence. Then, we applied PCA to reduce to 50-dimensional vectors, keeping the main information while removing noise.
- TF–IDF + SVD (100 dimensions): we applied TF–IDF using only 1-grams and 2-grams (up to 10,000 features), then reduced the resulting vectors to 100 dimensions with TruncatedSVD.

2.2 Analysis and Training on the Data

2.2.1 Clustering: dividing into groups of users

We hypothesized that semantically similar users form specific groups. We applied K-Means (12 clusters) on the PCA-reduced RoBERTa embeddings across train and test sets. Cluster assignments were one-hot encoded and added as new features.

2.2.2 Modeling Approach

Furthermore, we trained a stacked ensemble of three gradient boosting models:

- LightGBM: uses strong L1/L2 regularization and keeps trees shallow.
- XGBoost: builds deeper trees with higher regularization.

– CatBoost: handles mixed numeric and categorical features robustly.

The optimal weights were found using scipy.optimize and the final ensemble prediction of these three models was computed by:

$$p = \frac{1}{3}\, p_{\text{LGBM}} + \frac{1}{3}\, p_{\text{XGB}} + \frac{1}{3}\, p_{\text{CatBoost}}$$

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The optimal validation threshold was empirically determined as 0.515.

2.3 Avoid Overfitting and Safe Pseudo-Labelling

Overfitting was mitigated using a strong regularization, feature subsampling, early stopping and PCA/SVD dimensionality reduction.

Moreover, we used pseudo-labelling by introducing high-confidence predictions ($p > 0.95$ or $p < 0.05$) in from the test set in the train set. These represented roughly 8–12% of test samples and retraining the ensemble produced a small imporvement in validation accuracy.

Conclusion

We developed a hybrid architecture combining NLP embeddings, clustering, and behavioural feature engineering. Despite predicting from a single tweet, the model captures latent behavioural signatures distinguishing Influencers from Observers. Future improvements may include transformer fine-tuning, temporal modelling of user behaviour, or graph-based analysis of user interactions. The pipeline is fully reproducible and achieved a competitive final score of 0.843 accuracy.